

Lab 0 – Python exercises

DAVE3625 Introduction to Artificial Intelligence · Høst 2026

Solve these in `lab0.ipynb`. Compare with [Solution.ipynb](#) when you are done.

Task 1: "Hello World!"

Write and run a program that prints `Hello World!`.

Task 2: Basic arithmetic

Perform basic arithmetic operations in Python.

1. Write code to:
 - Add two numbers.
 - Multiply two numbers.
 - Divide one number by another.
2. Display the results of each calculation.

Task 3: Variables

Learn to store and use values in variables.

1. Create variables to store a name and an age.
2. Print the values of the variables in a formatted string.

Task 4: Simple loops

Use a loop to repeat an action multiple times.

1. Write a for-loop that iterates over a range of numbers.
2. In each iteration, print a line of text that includes the current iteration number.

Task 5: Conditional statements

Implement decision-making in your code using if-else statements.

1. Write an if-statement that checks if a number is greater than 5.
2. Print a message based on whether the condition is true or false.

Task 6: Match-case statements (*needs Python 3.10+*)

Learn to use Python's modern match-case syntax for pattern matching (Python 3.10+).

1. Create a variable holding a day of the week as a string, e.g. `"Monday"`.
2. Use a `match / case` statement to print:
 - `"Start of the work week!"` for Monday
 - `"Midweek already!"` for Wednesday
 - `"Almost weekend!"` for Friday
 - `"Weekend time!"` for Saturday or Sunday
 - `"Just another day"` for anything else

3. Test it with different days.

Bonus: make it handle both lowercase and uppercase input, e.g. `"monday"` and `"Monday"`.